

21CSC529T Inferential Statistics

Unit - 2: PROBABILITY

Unit 2

- Permutation and Combination
- Types of Probability
- Rules for computing probability
- Marginal probability
- Conditional probability
- Bayes Theorem
- Applications of Bayes Theorem
- Problem-solving in Bayes theorem
- Probability Distributions Binomial & Poisson - Normal Distribution

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Probability Distributions

Agenda

- Random Variable
 - Discrete Random Variable
 - Continuous Random Variable
- Probability Distribution
 - Discrete Probability Distribution
 - Binomial Distribution
 - Poisson Distribution
 - Cumulative Distribution Function
 - Continuous Probability Distribution
 - Normal Distribution

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Statistics



Random Variable

Random variable

- The action of rolling a die is called a **random experiment**
- The set of possible outcomes of an experiment is called the **Sample Space** and it is denoted by Ω
- Possible outcomes for this experiment are: 1, 2, 3, 4, 5, and 6
- A **Random Variable** is a function defined from the sample space to real numbers

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Random variable

- The usual notation used for a random variable is 'X'. However, other capital letters like U, V, or Z can be used
- Let us observe the experiment of number appeared on the die. We define our random variable as

X: the number observed

X takes values 1, 2, 3, 4, 5, and 6.

Hence $\Omega = \{1, 2, 3, 4, 5, 6\}$

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Random variable

Question:

Suppose you are the owner of a grocery shop. You wish to improve the shopping experience of your customers, by easing their payment methods and ensuring that whichever products they demand are available in the shop.

Now collect the data for:

- mode of payment
- bill amount

Can you describe the nature of the data collected?



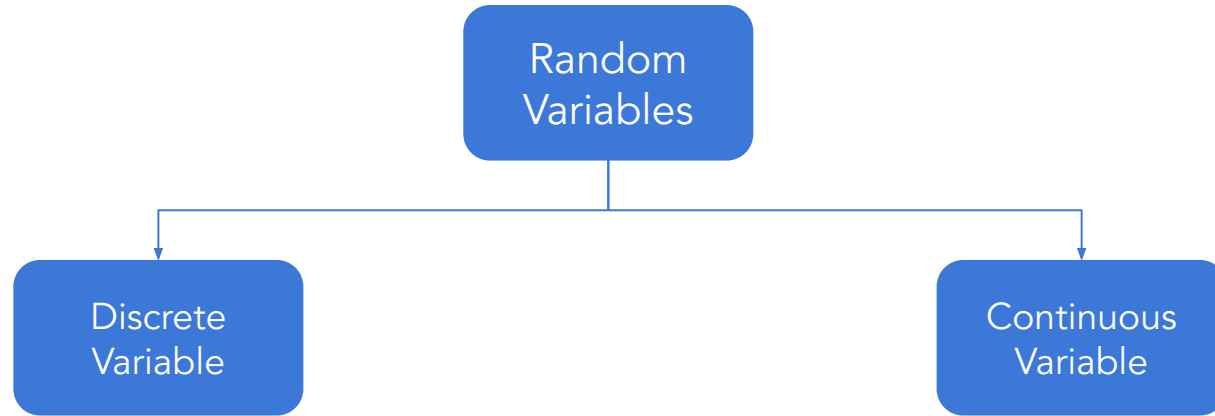
Random variable

Solution:

The nature of the data collected is as follows:

- mode of payment: the data takes two distinct values *Cashless payment* and *Cash payment*
- bill amount: It takes values such as \$11.45, \$32, \$8.

Random variable



Eg: The mode of
payment - cash, mobile
wallet, card, bank
transfer

Eg: The bill
amount

Discrete random variable

- A discrete random variable takes discrete values that is value from the set of whole numbers only or a set of categories
- A discrete random variable can either take finite or countably infinite values
- The possible outcomes can be listed effectively

Discrete random variable

Examples

- Let V : The number of pages in a document
 $\Omega = \{1, 2, 3, \dots\}$
- Let X : The number of views for a youtube video
 $\Omega = \{0, 1, 2, 3, \dots\}$
- Let U : The size of the shirt available
 $\Omega = \{XS, S, M, L, XL\}$

Continuous random variable

- A continuous random variable takes infinitely many values (within a specified range)
- It can either take an uncountably infinite set of values or values in a given interval

Continuous random variable

Examples:

- Let Z : The income of a person
 $\Omega = (0, \text{maximum income of a person})$
- Let V : The level of water in a dam
 $\Omega = (0, \text{height of the dam})$

Note that random variables Z and V take values from a range and are thus continuous random variables

Identify whether the variable is discrete or continuous random variable.

1. Accidents on the New York highway
2. Fuel capacity of a car
3. Amount of milk in 15-ounce bottle
4. Members of the public health society
5. The total duration of a movie
6. The number of movies released in a month

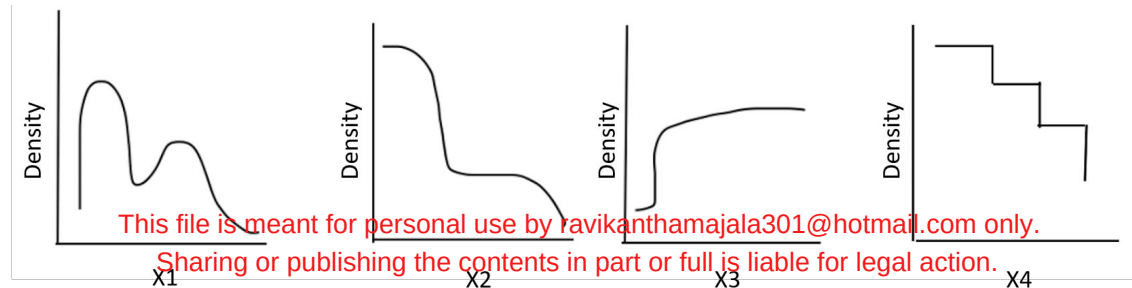
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Probability Distributions

Distribution of the data

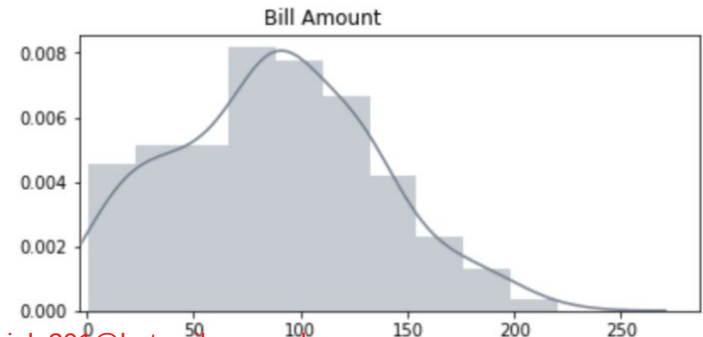
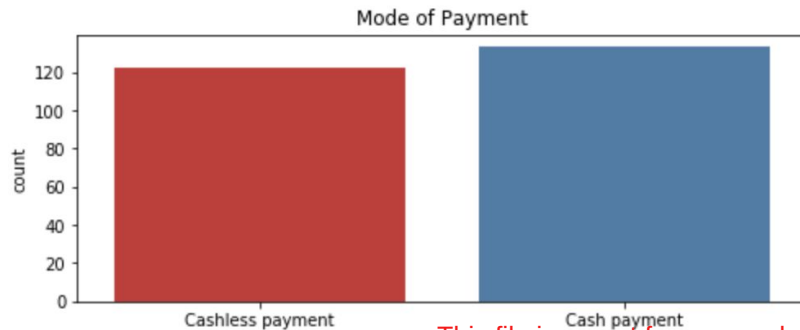
- The distribution is a summary of the frequency of values taken by a random variable
- The distribution of the data gives information on the shape and spread of the data
- On plotting the histogram or a frequency curve for a variable, we actually look at how the data is distributed over its range



Probability distribution

Recall the example of data collection at the store. The data was collected for the mode of payment and the bill amount.

If we represent the data will have the following figures:



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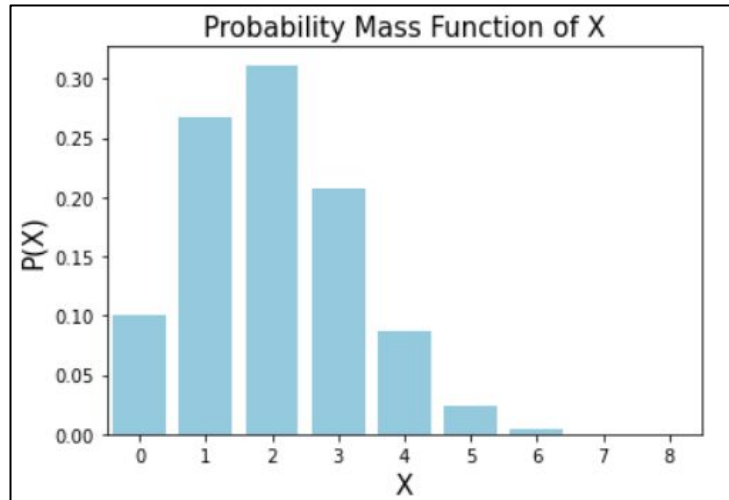
Need for probability distribution

- The statistical tests have assumptions based on the probability distributions that are needed to be verified before using the test
- Once the distribution of the data is known, the descriptive statistics of the data is readily available
- For instance, if the stock trader can determine the probability distribution of a particular stock, then the trader can estimate the potential expected returns that may yield in the future

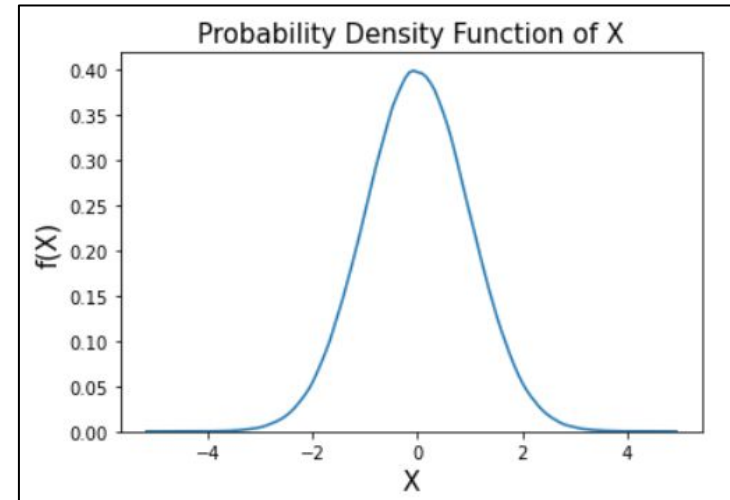
Probability distribution

- A probability distribution is a list of all the possible outcomes of a random variable along with their corresponding probability of occurrence
- $P(X=x)$ refers to the probability that the random variable X takes a particular value x
- A probability distribution is defined by its
 - probability mass function (p.m.f) for a discrete random variable
 - probability density function (p.d.f) for a continuous random variable

Probability distribution



Discrete Variable



Continuous Variable

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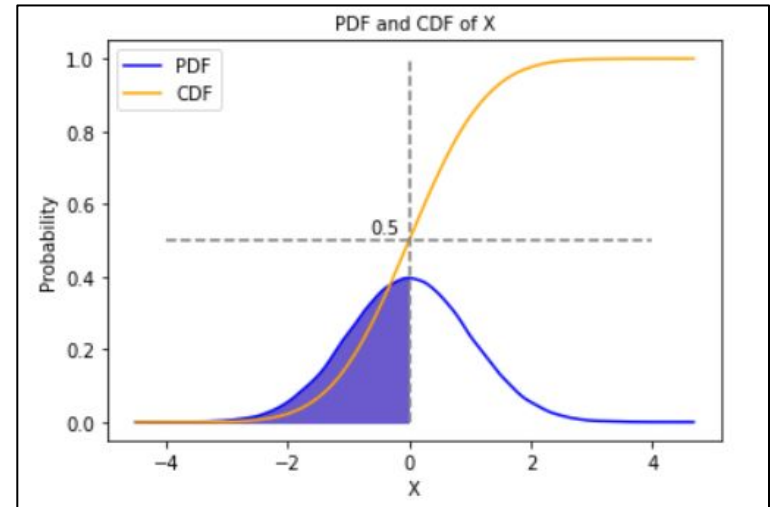
The PMF and PDF difference

The mathematical difference between a discrete distribution and a continuous distribution is:

- The discrete distribution is defined by the probability mass function (pmf) which is the probability at a particular point
- The continuous distribution is defined by the probability density function (pdf) which is the **integration** over it's range

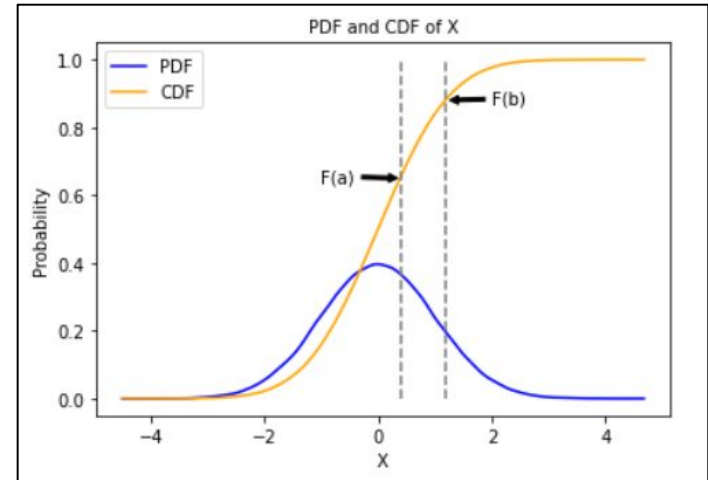
Cumulative distribution function (c.d.f.)

- It gives the cumulative distribution function i.e $P(X \leq x)$
- Denoted by $F(x)$
- That is, $F(x) = P(X \leq x)$
- The area of shaded region is 0.5



Properties of c.d.f.

- $0 \leq F(x) \leq 1$ since it is a probability
- An increasing function or monotonically non-decreasing function
- $F(-\infty) = 0$ and $F(\infty) = 1$
- For any two numbers a and b such that $a < b$, the probability is computed as:



$$P(a \leq X \leq b) = F(b) - F(a)$$

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Mathematical expectation of a random variable

- It is the mathematical mean of a random variable
- It is the central location
- Also known as **expected value**
- Denoted by μ
- For a probability distribution, it is $\mu = \sum x \cdot f(x)$

Where $f(x)$ is the probability mass function

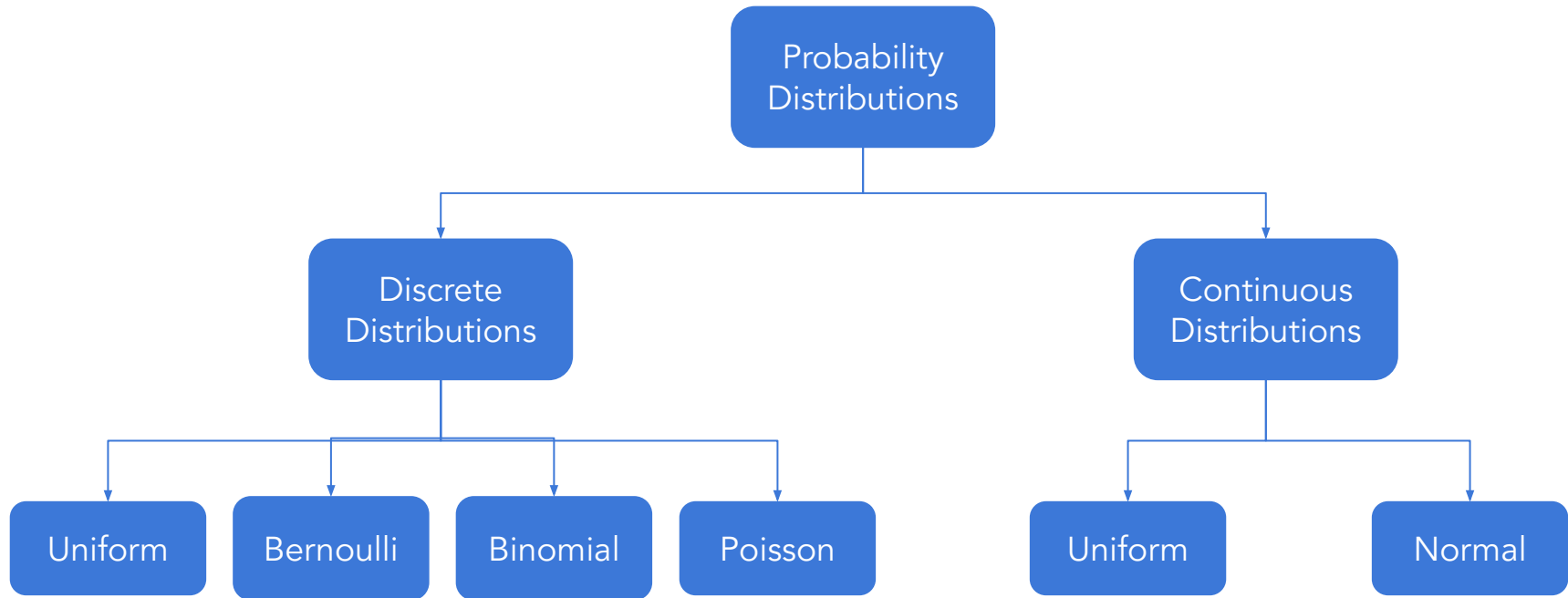
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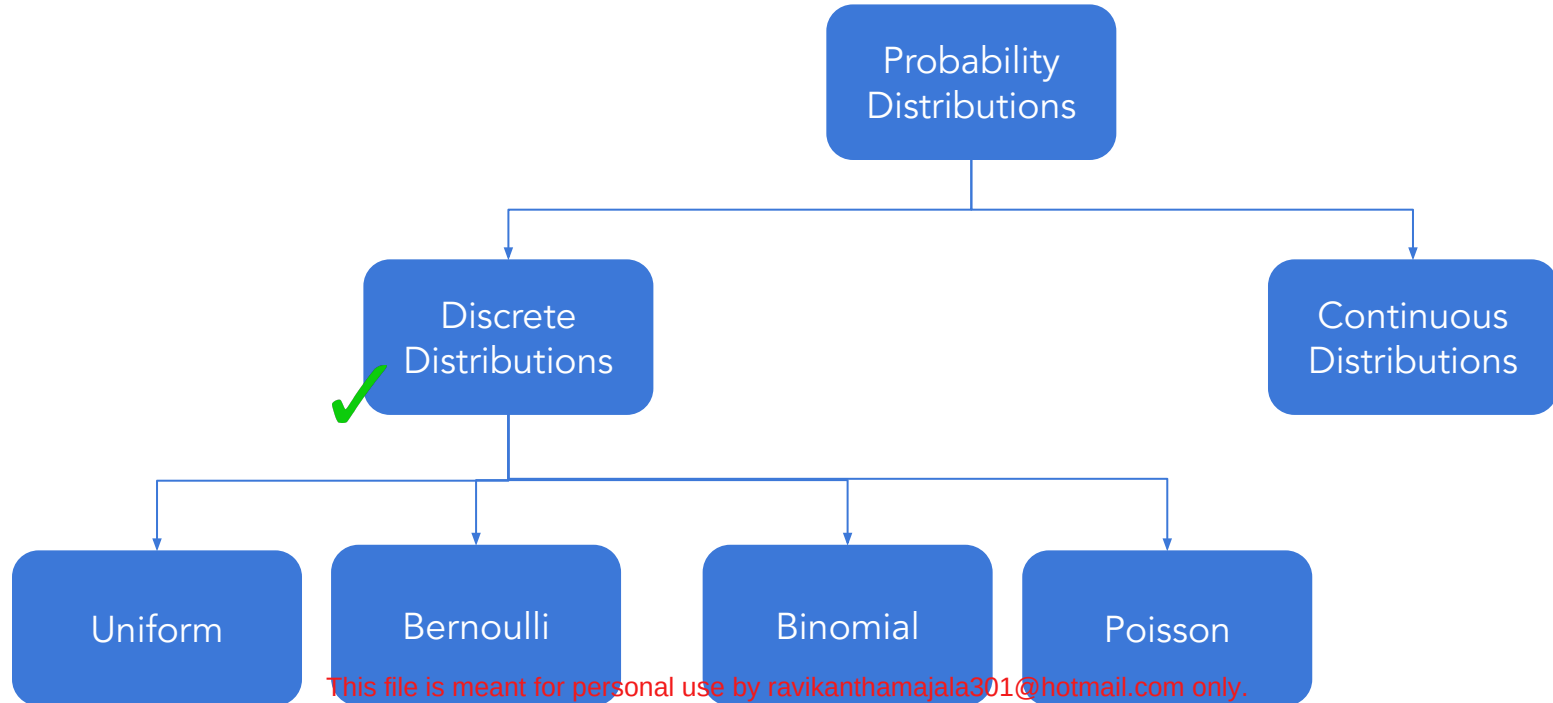
Variance of a random variable

- It is the mathematical mean of squares of deviations taken from the mean
- Denoted by σ^2
- For a probability distribution, it is $\sigma^2 = \sum (x - \mu)^2 \cdot f(x)$
where $f(x)$ is the probability mass function

Probability distributions



Probability distributions



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Discrete probability distribution

- A discrete distribution comprises of the probabilities of the outcomes of a random variable with **finite values** and is used to model a discrete random variable
- A discrete probability distribution is defined by its probability mass function (p.m.f)
- The mathematical expectation $E(X)$ and variance $V(X)$ of a discrete probability distribution is given by

$$E(X) = \sum_i x_i \cdot P(X = x_i)$$

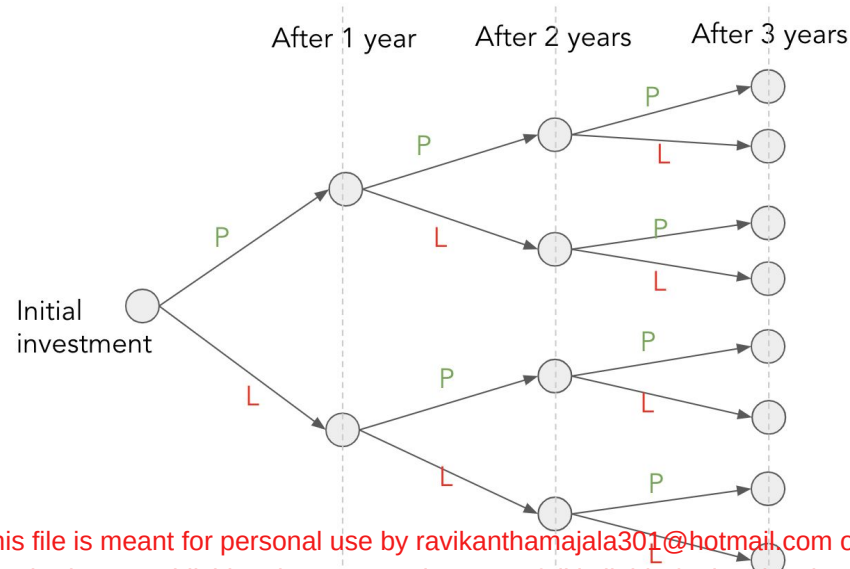
$$Var(X) = E[(X - \mu)^2] = \sum_i (x_i - \mu)^2 \cdot P(X = x_i)$$

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Binomial distribution

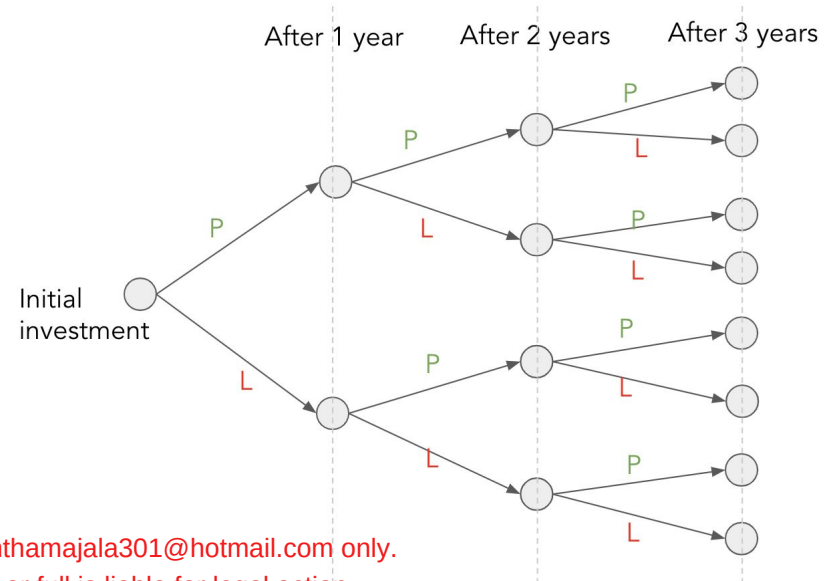
Steve has invested in a scheme. The following tree shows all the possibilities of how his investments may vary year to year over a period of three years



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Binomial distribution

- Let X be the event of a change in the initial invested amount
- There is a profit with probability p , naturally the loss incurred will have the probability $1-p = q$ (say)
- Let the probability of profit be 0.4
i.e. $P(X = P) = p = 0.4$
- Let the probability of loss be 0.6
i.e. $P(X = L) = 1 - p = 1 - 0.4 = 0.6$



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Binomial distribution

We can easily tabulate these probabilities.

Combination	Number of Profits	Probability of the combination
PPP	3	$0.4 \times 0.4 \times 0.4 = 0.064$
PPL	2	$0.4 \times 0.4 \times 0.6 = 0.096$
PLP	2	$0.4 \times 0.6 \times 0.4 = 0.096$
PLL	1	$0.4 \times 0.6 \times 0.6 = 0.144$
LPP	2	$0.6 \times 0.4 \times 0.4 = 0.096$
LPL	1	$0.6 \times 0.4 \times 0.6 = 0.144$
LLP	1	$0.6 \times 0.6 \times 0.4 = 0.144$
LLL	0	$0.6 \times 0.6 \times 0.6 = 0.216$
Total probability		1

Number of Profits	Probability
0	0.216
1	0.144
2	0.096
3	0.064

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Binomial distribution

- It is based on the Bernoulli distribution
- A discrete random variable (X) taking values 0,1,2,...,n follows a binomial distribution if the p.m.f. of X is given as:

$$P(x) = \binom{n}{x} p^x q^{n-x} \quad x = 0, 1, 2, \dots, n$$
$$= 0 \quad \text{otherwise}$$

- X follows binomial distribution with parameters n and p , i.e $X \sim \text{Binomial}(n, p)$
- p is the probability of success, q is the probability of failure and n is the number of times the experiment is conducted

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Binomial distribution

Recall Steve's example, multiply each probability with nC_x

Number of Profits (x)	Probability	3C_x	$P(x) = {}^3C_x \text{ Probability}$
0	0.216	${}^3C_0 = 1$	0.216
1	0.144	${}^3C_1 = 3$	0.432
2	0.096	${}^3C_2 = 3$	0.288
3	0.064	${}^3C_3 = 1$	0.064

$$P(x) = \binom{n}{x} p^x q^{n-x}$$

$$= 0$$

$x = 0, 1, 2, \dots, n$
otherwise

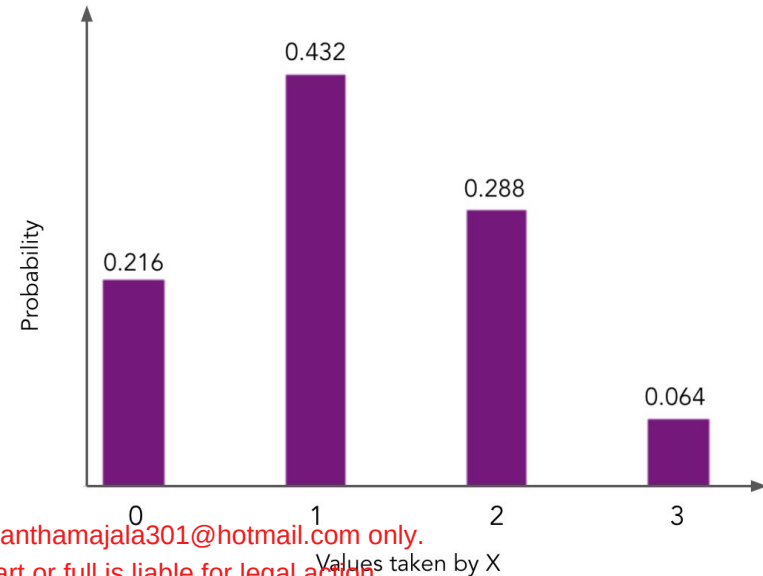
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Binomial distribution

The probability of the Binomial(3, 0.4)

Number of Profits (x)	$P(x) = {}^3C_x$ Probability
0	0.216
1	0.432
2	0.288
3	0.064



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If n independent random variables are Bernoulli distributed, then the sum of these random variables follows a Binomial distribution

Binomial distribution

The mean and the variance of the distribution is given as:

$$\begin{aligned} \text{Mean} = E(X) &= \sum_{x=0}^n xP(x) \\ &= \sum_{x=0}^n x \binom{n}{x} p^x q^{n-x} \\ &= np \end{aligned}$$

$$\begin{aligned} \text{Variance} &= E(X^2) - [E(X)]^2 \\ &= npq \end{aligned}$$

For our example, if $n = 10$, $p = \frac{2}{3}$ and $q = \frac{1}{3}$. Then, $E(X) = \frac{20}{3}$ and $V(X) = \frac{20}{9}$

Binomial distribution - python function

Python function	Description
<code>scipy.stats.binomial.pmf()</code>	returns the pmf of discrete binomial distribution
<code>scipy.stats.binomial.cdf()</code>	returns the cdf of discrete binomial distribution
<code>scipy.stats.binomial.sf()</code>	calculates the value of survival function ($1 - \text{cdf}$)



Binomial distribution

Question:

In a shooting academy, data was collected on the precision shooting of a student. From 15 shots fired 11 were on target. Considering the same student, what is the probability that out of 50 shots fired, exactly 35 will hit the target?



Binomial distribution

Solution:

Let X denote the shot fired hits the target

The probability of success, i.e. is hitting the target is $11/15$

Thus, $p = 0.733$

To find: the probability that out of 50 shots fired, 35 will hit the target

Here, $n = 50$

We have $X \sim \text{Binomial}(50, 0.733)$

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Binomial distribution

Solution:

The required probability is $P(X = 35)$

$$P(X = x) = {}^nC_x p^x (1-p)^{n-x}$$

$$\begin{aligned} P(X = 35) &= {}^{50}C_{35} (0.733)^{35} (1-0.733)^{50-35} \\ &= {}^{50}C_{35} (0.733)^{35} (0.267)^{15} \\ &= 0.107 \end{aligned}$$

Thus, the probability that out of 50 shots fired, exactly 35 will hit the target is 0.107

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Binomial distribution

Python solution:

```
# use 'binom.pmf()' to calculate the pmf for binomial distribution
# pass the required value of shots hit on the target to the parameter, 'k'
# pass number of total shots fired to the parameter, 'n'
# here the success is hitting the shots on the target with probability 11/15
prob = stats.binom.pmf(k = 35, n = 50, p = 11/15)

# use 'round()' to round-off the value to 3 digits
prob = round(prob, 3)
print('The probability that that out of 50 shots fired, exactly 35 will hit the target is', prob)
```

The probability that that out of 50 shots fired, exactly 35 will hit the target is 0.107

Poisson distribution

- It is used to determine the probability of events that occur in a fixed interval of time/space
- It is used to describe the distribution of rare events in a large population
- A discrete random variable X follows a Poisson distribution with parameter λ , if p.m.f. of X is given as:

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!} \quad \begin{matrix} x = 0, 1, 2, \dots \\ \lambda > 0 \end{matrix}$$

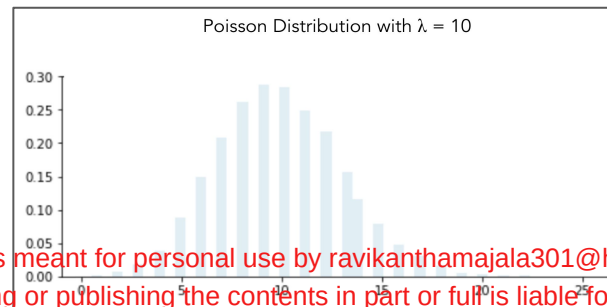
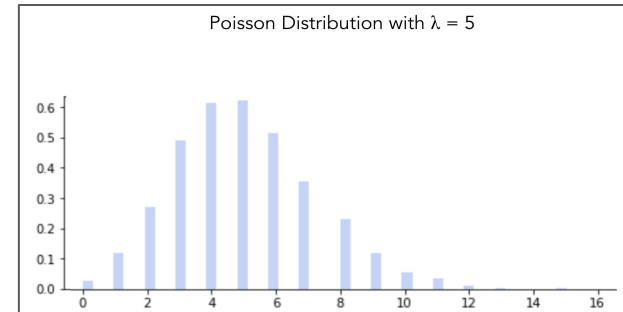
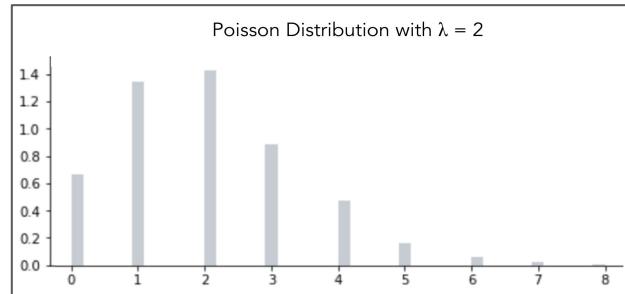
Where,

e = Euler's number (= 2.718)

$x!$ = Factorial of x

Poisson distribution

The Poisson distribution is positively skewed.



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Poisson distribution

The mean and the variance of the distribution is given as:

$$\begin{aligned} \text{Mean} = E(X) &= \sum_{x=0}^{\infty} xP(x) \\ &= \lambda \end{aligned}$$

$$\begin{aligned} \text{Variance} &= E(X^2) - [E(X)]^2 \\ &= \lambda \end{aligned}$$

Mean and variance of the poisson distribution are equal and it is the parameter of distribution.

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**WANT
TO KNOW
MORE?**

Poisson as a limiting case of binomial distribution

Let $X \sim \text{Binomial}(n, p)$ such that for large n ($n \rightarrow \infty$) and small p , $np = m$ is a constant then X approaches $\text{Poisson}(m)$

Poisson distribution - python function

Python function	Description
<code>scipy.stats.poisson.pmf()</code>	returns the pmf of discrete poisson distribution
<code>scipy.stats.poisson.cdf()</code>	returns the cdf of discrete poisson distribution
<code>scipy.stats.poisson.sf()</code>	calculates the value of survival function (1 - cdf)



Poisson distribution

Question:

A call center receives an average of 15 calls per hour. The arrival of calls is random, and the distribution of the number of calls received per hour can be modeled as a Poisson process.

1. What is the probability that the call center will receive exactly 20 calls in a given hour?



Poisson distribution

Solution:

Let X: number of calls received by call center

The average number of calls per hour is **15**, so the Poisson parameter for this problem is: $\lambda=15$ calls/hour

We know $P(X=x) = f(\lambda)$

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$$



Poisson distribution

Solution:

Probability of Receiving Exactly 20 Calls in an Hour

The probability mass function (PMF) of a Poisson distribution is given by:

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

We have,

$$P(X = 20) = \frac{15^{20} e^{-15}}{20!}$$



Poisson distribution

Solution:

We can calculate the components individually:

$$P(X = 20) = \frac{15^{20} \cdot e^{-15}}{20!}$$

$$P(X = 20) = \frac{3.27 \times 10^{23} \cdot 3.06 \times 10^{-7}}{2.43 \times 10^{18}}$$

$$P(X = 20) = \frac{1.00 \times 10^{17}}{2.43 \times 10^{18}} \approx 0.0412$$

The probability that the call center will receive **exactly 20 calls in a given hour** is approximately **0.0412**, or **4.12%**

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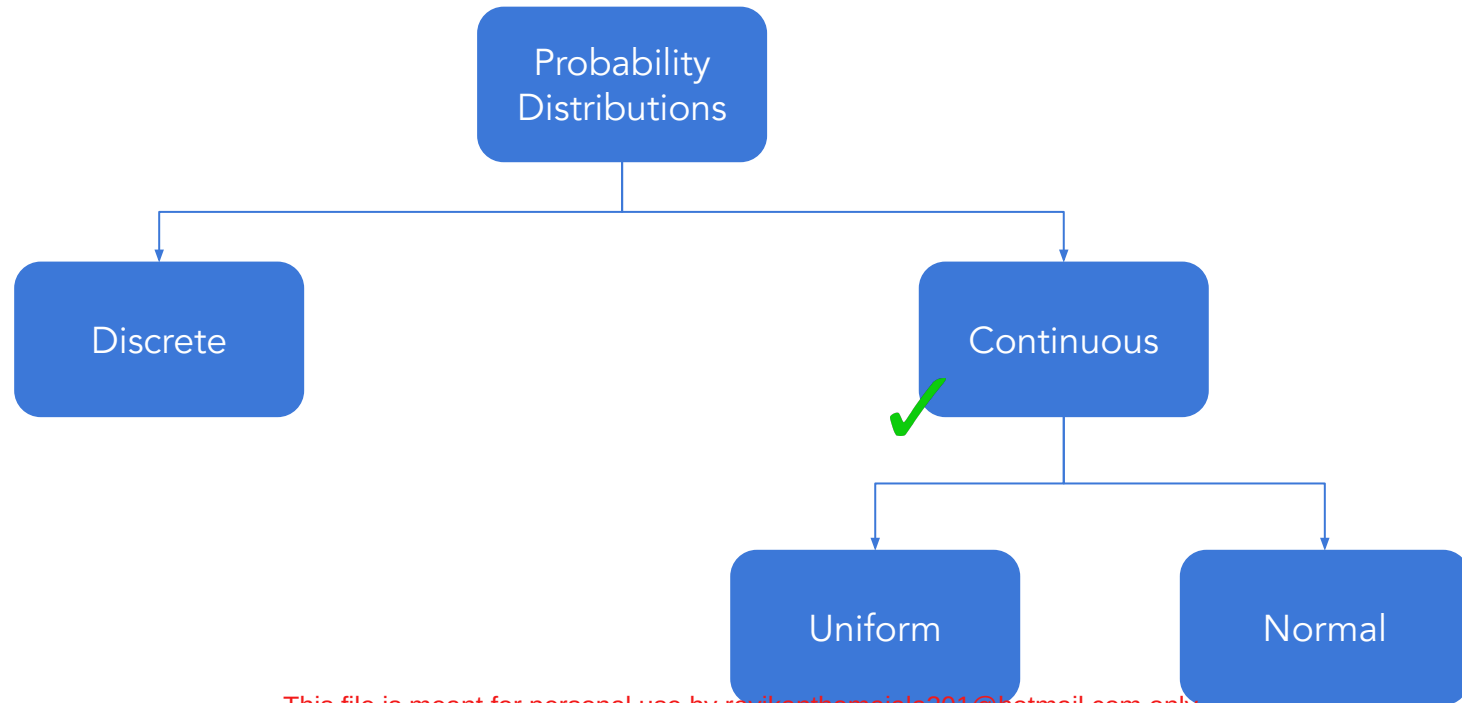


A local bakery receives an average of 12 customer orders per hour. The owner wants to determine the likelihood of receiving exactly 10 orders in a particular hour, as well as the probability of receiving no more than 5 orders during the same hour. Understanding these probabilities will help the bakery staff manage their inventory and staffing needs more effectively. Additionally, the owner is interested in the overall probability of receiving more than 30 orders during the busiest part of the day, which lasts for 3 hours.

Summary

- Some of the discrete distributions are: Uniform, Bernoulli, Binomial and Poisson
- Discrete distribution is determined using probability mass function (p.m.f.)
- In uniform distribution, all the values of the variable are equally likely
- The sum of independent and identically distributed Bernoulli variables follows binomial distribution
- Poisson distribution can be considered as a limiting case of a binomial distribution

Probability distributions



Continuous probability distribution

- It is a probability distribution of the continuous random variable
- The continuous distributions are defined by their probability density function (p.d.f.) denoted as $f(x)$
- The probability is given by the area under the pdf on a specified range. Thus, the value of the probability density function is 0 at a particular point
- The area under the curve is always equal to 1. Therefore, is considered as a probability distribution

Normal distribution

The probability density function for the normal distribution is,

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

It is said that $X \sim N(\mu, \sigma^2)$, i.e the variable X follows the normal distribution with parameter μ and σ^2

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Normal distribution

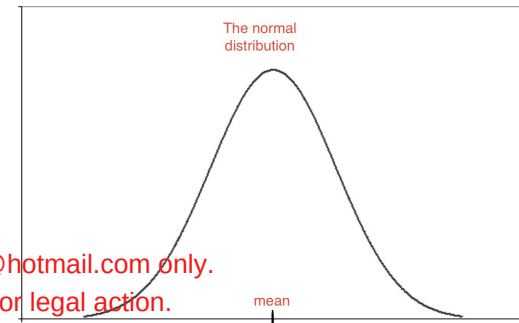
The mean and the variance of the distribution is given as:

$$\begin{aligned} \text{Mean} = E(X) &= \int_{-\infty}^{\infty} x f(x) dx \\ &= \mu \end{aligned}$$

$$\begin{aligned} V(X) &= E(X) - [E(X)]^2 \\ &= \sigma^2 \end{aligned}$$

Normal distribution

- Also called as the Gaussian distribution
- The frequent observations are found in the middle of the distribution, and further decrease away towards the tails
- A normal distribution has a bell-shaped density curve described by its mean and standard deviation
- For normally distributed data,
 - Mean = Mode = Median
 - Symmetric about the center



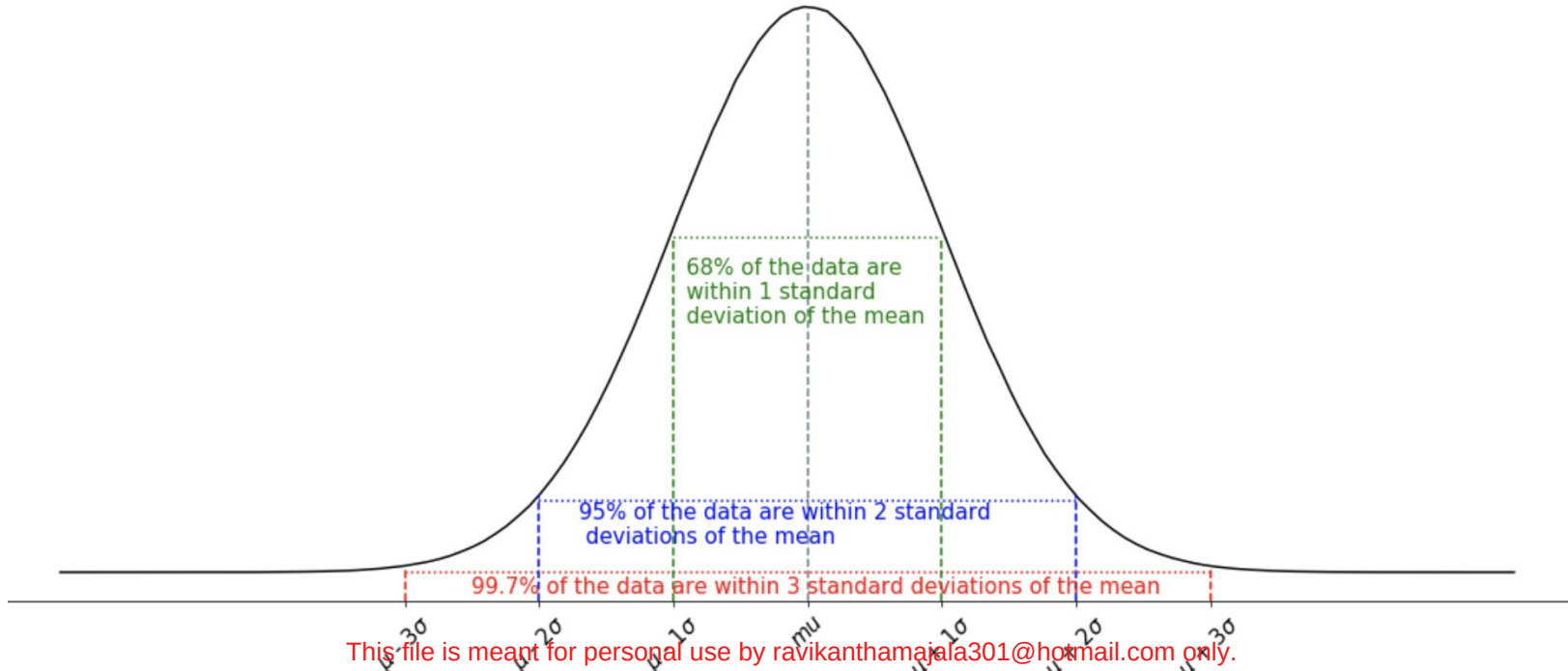
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Characteristics of normal distribution

- The distribution is symmetric about the mean
- For $X \sim N(\mu, \sigma^2)$, $P(X > a) = P(X < -a)$ due to symmetry
- The standard normal distribution has mean 0 and variance 1
- For $X \sim N(\mu, \sigma^2)$, if $Z = (x - \mu)/\sigma$ then $Z \sim N(0, 1)$ that is the standard normal distribution

The normal distribution spread



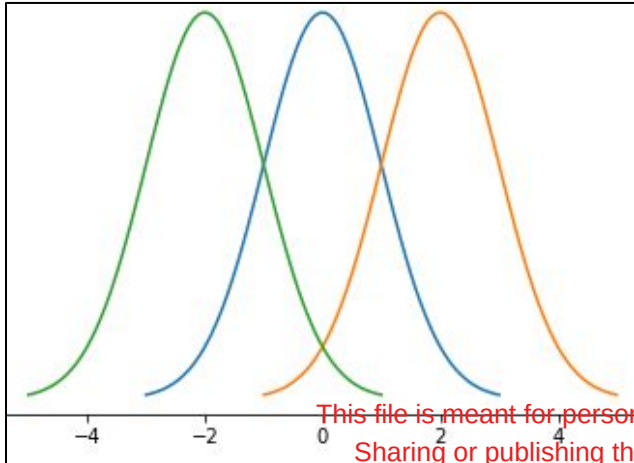
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Normal distribution

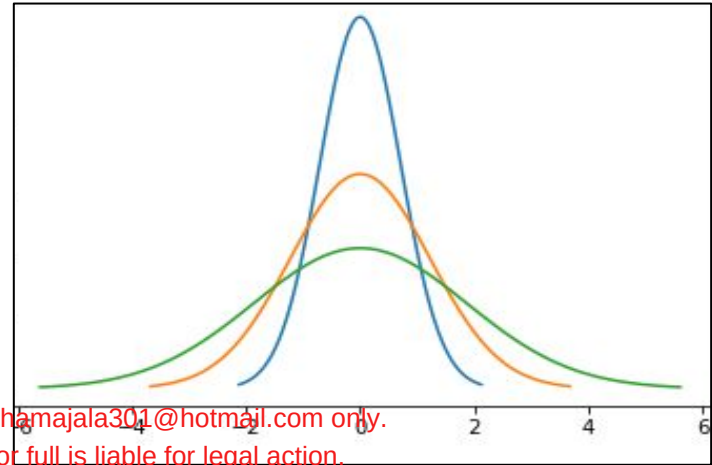
- IQ of the human population is the example of normal distribution
- Most of the people in the specific population have average IQ
- The number of people with higher IQ and lower IQ than the average IQ is almost equal, and a very less number of people have extremely less IQ or extremely high IQ

Different shapes of normal distribution

Normal distributions with same standard deviation but different means:



Normal distributions with same mean but different standard deviations:



Normal distribution - python function

Python function	Description
<code>scipy.stats.normal.pmf()</code>	returns the pmf of discrete normal distribution
<code>scipy.stats.normal.cdf()</code>	returns the cdf of discrete normal distribution
<code>scipy.stats.normal.sf()</code>	calculates the value of survival function (1 - cdf)



Normal distribution

Question:

A monthly balance in the bank account of credit card holders is assumed to be normally distributed with mean \$500 and variance \$100. What is the probability that the balance can be more than \$513.5?



Normal distribution

Solution:

Let X: monthly balance in the bank account of a credit card holders

$$X \sim N(\mu = 500, \sigma^2 = 100)$$

To find: the probability that the balance can be more than \$513.5

The required probability is $P(X > 513.5)$



Normal distribution

Solution:

$$\begin{aligned}P(X > 513.5) &= \int_{513.5}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{\sigma^2}} dx \\&= \int_{513.5}^{\infty} \frac{1}{\sqrt{2\pi \times 100}} e^{-\frac{(x-500)^2}{100}} dx \\&= \int_{513.5}^{\infty} \frac{1}{25.07} e^{-\frac{(x-500)^2}{100}} dx\end{aligned}$$

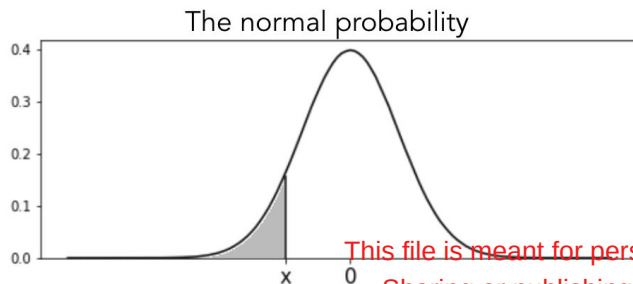
... Here is a problem. We see it is difficult to solve the integral, so we make use of the normal tables.

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The normal table

- Let the r.v. X strictly follow the standard normal distribution
- Let r.v. X take a negative value x
- This table gives the lower tail probabilities; $P(X \leq x)$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.6	.0002	.0002	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
-3.5	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

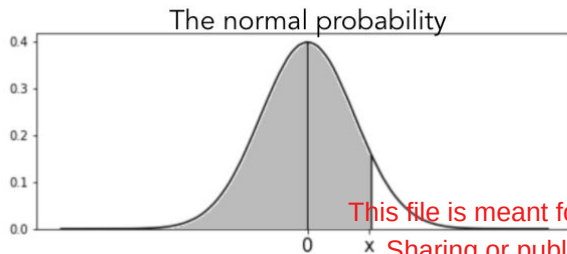
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The normal table

- Let the r.v. X strictly follow the standard normal distribution
- Let r.v. X take a positive value x
- This table gives the lower tail probabilities; $P(X \leq x)$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998
3.5	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998	.9998
3.6	.9998	.9998	.9999	.9999	.9999	.9999	.9999	.9999	.9999	.9999

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Normal distribution

Solution:

We know that for $X \sim N(\mu, \sigma^2)$, if $Z = (x - \mu)/\sigma$ then $Z \sim N(0, 1)$ that is the standard normal distribution

$X \sim N(\mu = 500, \sigma^2 = 100)$ thus $Z = (x - 500)/\sqrt{100} \sim N(0, 1)$

The required probability is $P(X > 513.5)$, now becomes

$$P(X > 513.5) = P\left(\frac{X - \mu}{\sigma} > \frac{513.5 - 500}{\sqrt{100}}\right) = P\left(Z > \frac{513.5 - 500}{\sqrt{100}}\right) = P(Z > 1.35)$$

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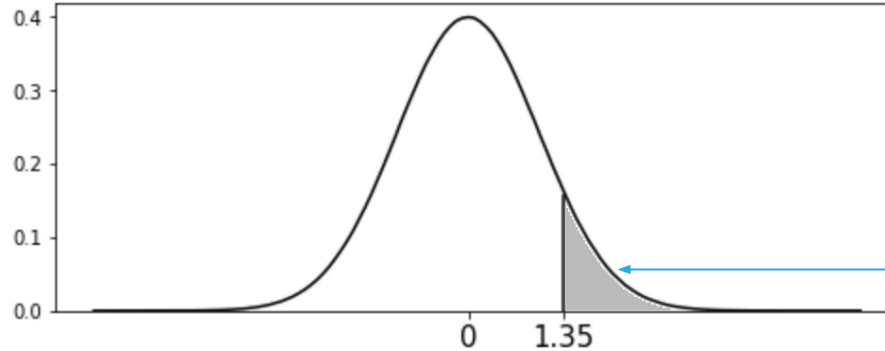
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Normal distribution

Solution:

Thus the required probability now is $P(Z > 1.35)$



The shaded region represents the required probability

Now we refer to the normal table to the probability

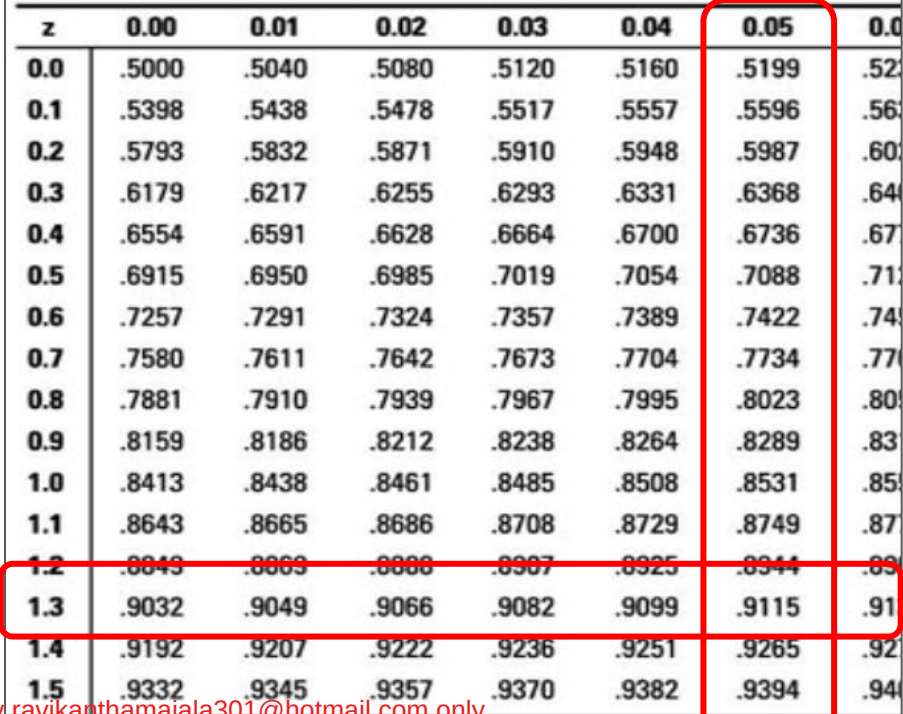


Normal distribution

Solution:

The table gives $P(Z \leq 1.35)$.

1. Look horizontally in the row of 1.3
2. Consider the value corresponding column to 0.05



2

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7122
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406



Normal distribution

Solution:

Since we get the lower tail probability as 0.9115

The required probability $P(Z > 1.35) = 1 - P(Z \leq 1.35)$

$$= 1 - 0.9115$$

$$= 0.0885$$

The probability that the balance can be more than \$513.5 is 0.0885



Normal distribution

Python solution:

```
# average account balance
avg = 500

# variance is 100 dollars
var = 100

# standard deviation is square-root of variance
std = np.sqrt(var)

# standardize the variable with x = 513.5
z = (513.5 - avg) / std

# calculate the probability that the balance is more than 513.5 dollars
# 'sf()' returns the P(Z > z) i.e P(Z > 513.5)
prob = stats.norm.sf(z)

# use 'round()' to round-off the value to 4 digits
req_prob = round(prob, 4)
print('The probability that the balance can be more than 513.5 dollars is', req_prob)
```

The probability that the balance can be more than 513.5 dollars is 0.0885

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The `scipy.stats.norm.cdf()` in python gives the lower tail probabilities

Summary

- Continuous distributions: Uniform and normal
- Continuous distribution is determined using probability density function (p.d.f.)
- The value of pdf at a particular point is always zero
- For a normal distribution, Mean = Median = Mode
- Standard normal distribution has mean zero and variance one

Thank you!

Happy Learning :)